1 Optical power is
a. light intensity per square centimeter.
b. the flow of light energy past a point.
c. a unique form of energy.
d. a constant quantity for each light source.

2. What measures power per unit area?
a. irradiance
b. intensity
c. average power
d. radiant flux
e. energy

3. A digitally modulated light source is on 25% of the time and off 75% of the time. Its rise and fall times are instantaneous. If its average power is 0.2 mW, what is the peak power?
a. 0.2 mW
b. 0.4 mW
c. 0.8 mW
d. 1.0 mW
e. impossible to calculate with the information given

on 25%
off 75%
average power is 0.2 mW } peak power?

P_{AVG} X (T_{ON} * T_{OFF} / T_{ON})
$0.2 * 10^{-3}$ X (0.25 * 0.75 / 0.25) = 0.8 mW

4. The light source in Problem 3 is left on for 10 s. How much energy does it deliver over that period?
a. 0.2 mW
b. 0.2 mj
c. 0.8 mj
d. 2 mj
e. 8 mj

Power = d (energy) / d (time)

A. **Find Energy**
   Energy = Power / time
   0.2 / 10 = 2 mJ

5. Light input to a 10-km long fiber is 1 mW. Light output at the end of the fiber is 0.5 mW. What is the fiber attenuation in dB/km?
a. 0.3 dB/km
b. 0.5 dB/km
c. 1 dB/km
d. 3 dB/km
e. 5 dB/km

A. Loss = $10\log_{10}(P_{OUT} / P_{IN})$
$10\log_{10}(0.5 * 10^{-3} / 1 * 10^{-3}) = 5$ dB
B. 5 dB / 10 km = 0.5 dB/km

6. Optical power is proportional to the
a. square of the optical intensity.
b. square of the optical energy.
c. wavelength times the speed of light.
d. square of the voltage applied to the detector.
e. square of the electric-field amplitude

9. A continuous 1-mW beam delivers light at 193.1 THz for one second. About how many photons is this equivalent to? Use a value of 6.626×10^{-34} J/Hz for Planck's constant h.
a. 10^{9} photons
b. 193.1×10^{9} photons
c. 7.82×10^{12} photons
d. 193.1×10^{12} photons
e. 7.82×10^{15} photons

A. $(6.626 * 10^{-34}$ J/Hz$) \times (193.1 * 10^{12}$ Hz$) = 1.2795 * 10^{-19}$J
B. 1-mW → $(1 * 10^{-3}$ J/s $/ 1.2795 * 10^{-19}$J$) = 7.82 * 10^{15}$ photons

10 . Pulse duration is
a. the interval between the time the rising pulse reaches half its maximum height to the time the falling pulse drops below that height.
b. the interval between successive peaks of the pulse.
c. the time it takes the pulse to rise from 10% to 90% of its maximum value.
d. half the cycle of a periodic sine wave.
e. the time from the start of one pulse to the start of the next.

11 . System bandwidth is measured as the
a. number of bits per second transmitted with no errors.
b. number of bits per second transmitted with a bit error rate of 10^{-12}
c. maximum frequency transmitted with no decline in power from lower frequencies.
d. frequency at which power has dropped 3 dB from the power at lower frequencies.
e. wavelength at which power has dropped 3 dB from the power at lower wavelengths.

12. What is the standard measure for transmission quality in digital systems?
a. signal-to-noise ratio
b. bit error rate
c. attenuation from transmitter to receiver
d. 3-dB bandwidth
e. pulse interval